

The main-dish line gets long before anyone reacts

In an IIT Guwahati hostel mess, the main-dish counter (dosa, paratha, roti) is the bottleneck. Staff only see the line once it is long; students cannot tell when to come.

86

people in line at the worst slot
(breakfast Saturday, 10:00) (real data)

17%

of 5-min intervals have a line of 10+,
so the problem is short and
predictable peaks (real data)

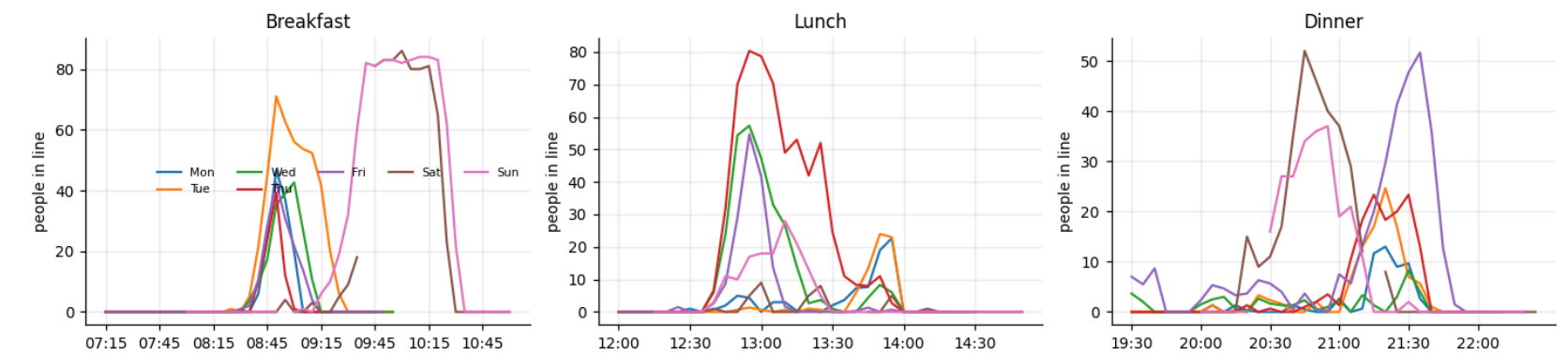
Users

- **Mess manager / counter staff:** how fast to serve in the next 15 minutes
- **Students:** go now, or come back in 15 minutes

Our question

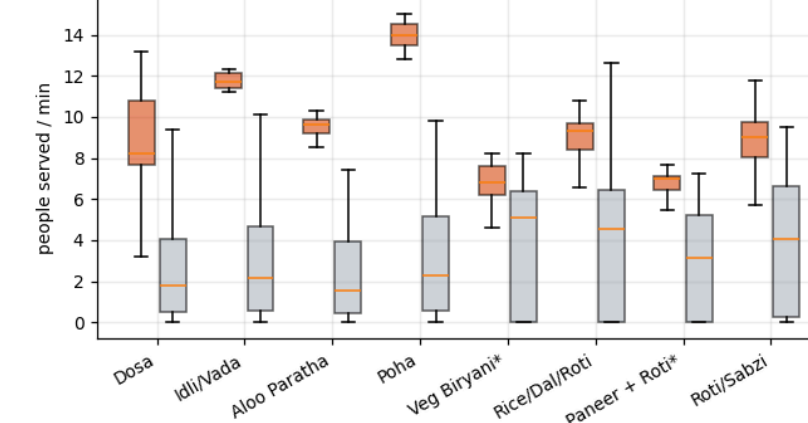
From door counts and 5-minute counter counts, **how long will the line be in 5, 10 and 15 minutes, how long is the wait, and what service rate keeps it under 5 minutes?**

Main-dish line length (mean by weekday)

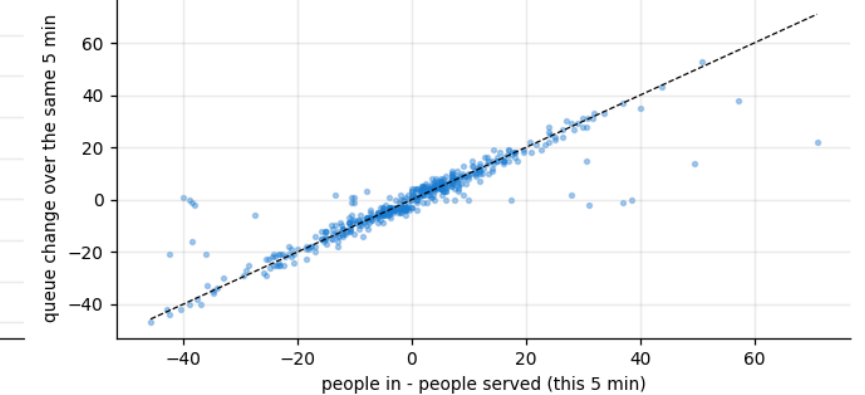


Mean line length by meal and weekday. Weekday rushes: breakfast 8:30-9:00, lunch 12:45-1:00 (Wed-Fri), dinner after 9:00; weekend breakfast burst at 9:45.

Service rate by dish: line present (orange) vs empty (grey)

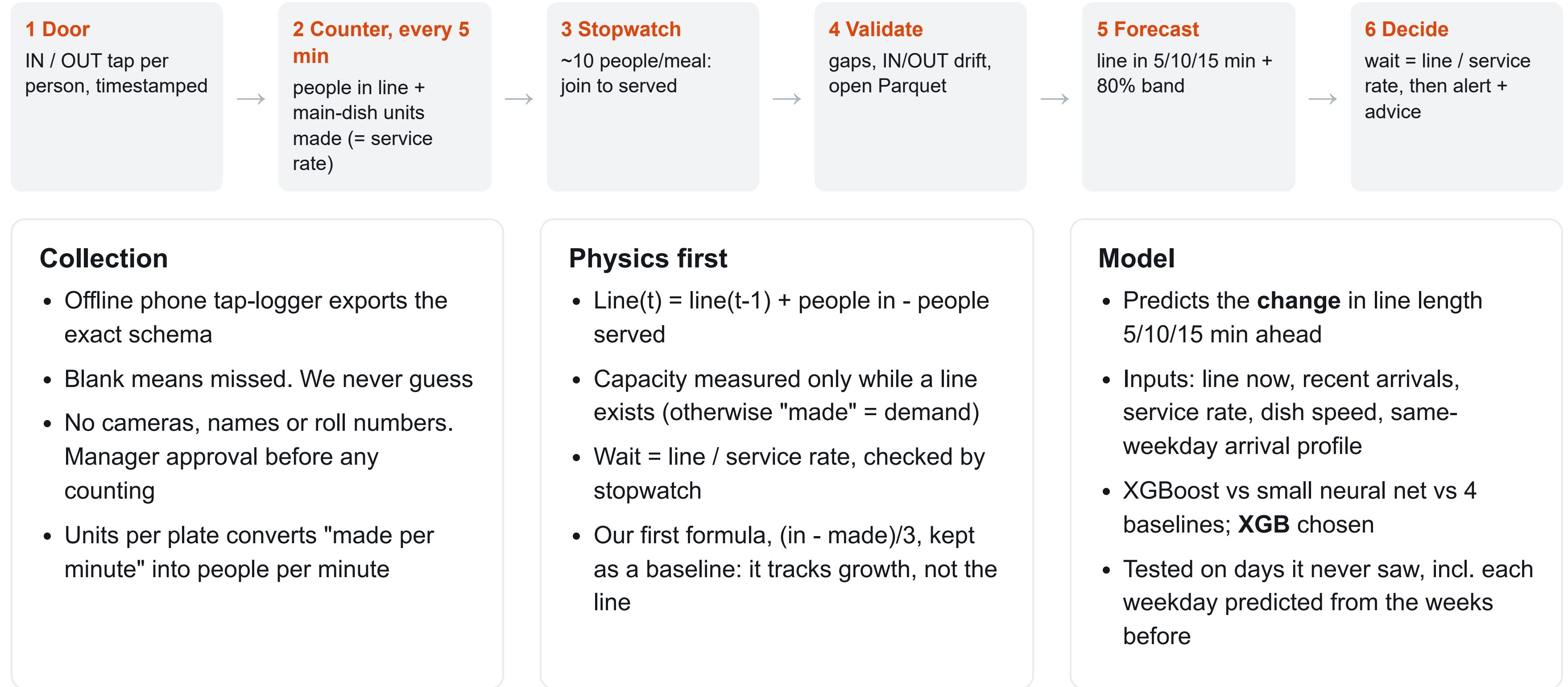


Queue balance holds when the line is present



Left: capacity depends on the dish. Right: line change = people in - people served.

Three phones, two counts, one forecast



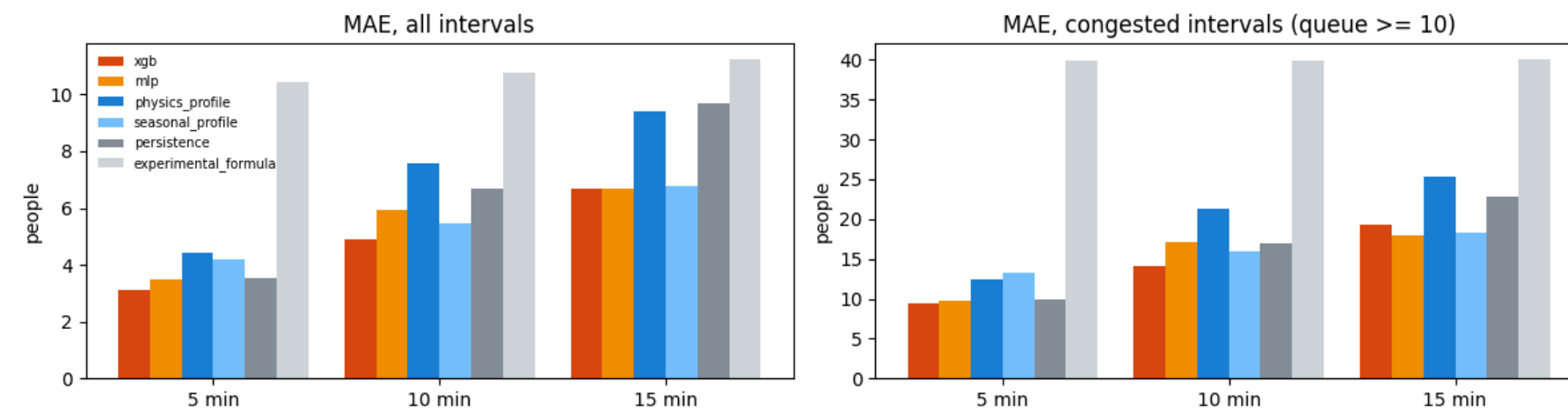
RESULT · WHAT CHANGES

A 15-minute warning the kitchen can act on

Held-out last 5 days

Congested-interval error, 10 min ahead: **14.1** people (XGB) vs 17.0 (line stays the same) and 39.9 (our first formula).

Forward Out-of-Sample Test (last 5 days held out)



Wait formula vs stopwatch (97 real people): error 0.3 min.

Manager card (held-out meal)

RED Dosa line: 83 now -> ~89 in 5 min (wait ~11 min)

Serving 8.1 people/min; need 20.1 (+148%) = 40 Dosa units/min

Action: add a server / cooking station or start the next batch now

09:45: line 83; 15 min later it was 83

What changes

- **Manager:** acts 10-15 minutes earlier with a target service rate, instead of reacting to a long line
- **Students:** "go now" or "come in 15 min"
- **After each meal:** peak, minutes over target, the service rate that was needed

Data Collection Challenges

- Tried accessing camera recordings but failed due to privacy issues and cameras not working.
- To resolve counting issues, we built an app to record: **people in** (entering mess and picking thali), **people out** (completed meal and returned thali), and **serving rate** (people leaving the queue).